LESSON PLAN · SCRATCH (GR 6–8) INTERMEDIATE: DATA & SIMULATIONS

Lesson 1: Lists and Traversal

Module 1 | 45 minutes | Lists and traversal | Grades 6–8 · ages 11–14

Objective: Store data in a list and traverse it with a loop and an index variable to process every item.

Success criteria

- I can traverse a list with an index.
- I can accumulate a total or count.
- I can spot off-by-one errors.

Key vocabulary

- **traverse**: Visit every item in a list.
- **index**: The position of an item (from 1 in Scratch).
- **accumulator**: A variable that builds a result in a loop.
- **off-by-one error**: A loop that starts or stops one step wrong.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)

Timed lesson plan (45 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Locker walk (5 min): numbered cards face down; a learner flips each in order, keeping a running total.
5 min	**Watch**	Play the lesson video with captions; pause at each heading and ask a question.
8 min	**Build along**	Weekly temperatures: learners follow the steps.
14 min	**Perform**	Code Lab: total the list, Off by one
9 min	**Probe**	Code Lab: count matching items, Trace in Python, activity
4 min	**Wrap-up**	Exit ticket: Which two blocks move the index and read the item?

Unplugged warm-up

Locker walk (5 min): numbered cards face down; a learner flips each in order, keeping a running total.

Teaching the idea

Traversal visits each list element using an index (1-based in Scratch) and a loop bounded by length. Accumulators (total, count) build results. Off-by-one errors are the classic bug. Python's for-in hides the index.

Analogy: Checking every locker in a row: start at locker 1, open it, move to the next.

Live demo: Trace the loop on the board with a table: i, item i, total.

Questions to ask

Ask	Listen for
Why repeat length of list times?	So it works for any size.
What does the accumulator start at?	0 for totals and counts.
What causes off-by-one errors?	Wrong start or end index.

Differentiation

Support
Provide a trace table template.

Stretch
Calculate the average: total / length of temps.

Common misconceptions

Misconception	How to fix it
Lists start at 0 in Scratch	They start at 1 (Python starts at 0).
repeat 7 is as good as repeat length	It breaks when the list size changes.

Assessment

Use the Code Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Which two blocks move the index and read the item?

Homework
Record a week of something (steps, temperatures, screen time) and plan how to total it with a loop.

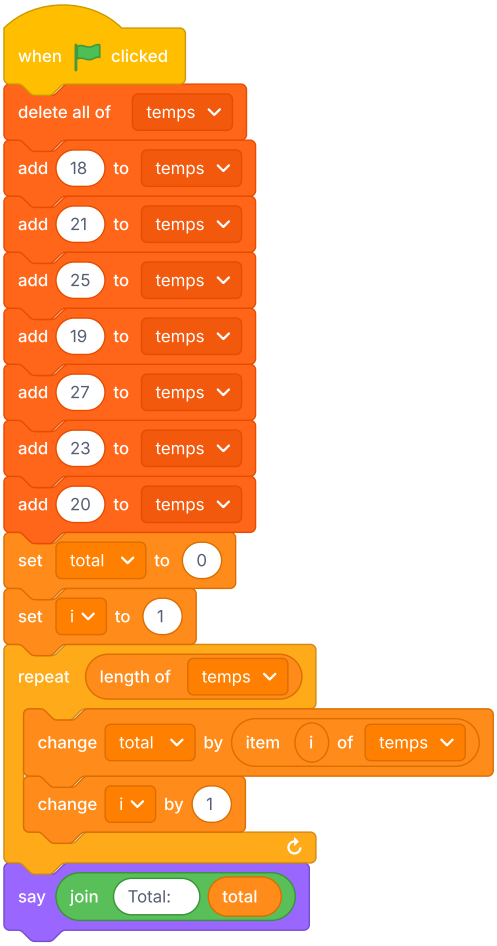
Answer key — Kitabash activities

1. Code Lab: total the list (perform · code lab)

Goals checked: Total is 153 · Uses an index · Loops over the list · Reports it

One solution:

Analyst



2. Off by one (perform · multiple choice)

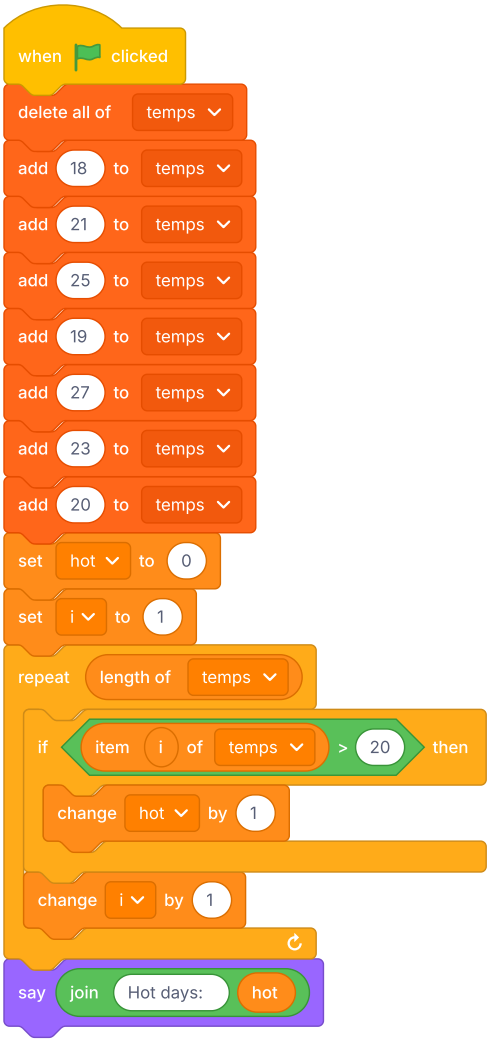
Answer: Item 0 doesn't exist, so the first read is empty and the last item is missed

3. Code Lab: count matching items (probe · code lab)

Goals checked: Counts 4 hot days · Uses a condition in the loop · Reports it

One solution:

Analyst



4. Trace in Python (probe · code output)

5. Explain your thinking (probe · free response)

Look for: index, item, loop. **Example:** The index i starts at 1 and the loop reads item i, then adds 1 to i so the next time round it reads the next item. If you forgot to change i, the loop would read item 1 every time and never see the rest of the list.

Answer key — worksheet

1. **Key words:** traverse = Visit every item in a list. · index = The position of an item (from 1 in Scratch). · accumulator = A variable that builds a result in a loop. · off-by-one error = A loop that starts or stops one step wrong.
2. **Plan your code:** Goals: Total is 153; Uses an index; Loops over the list; Reports it.

3. **Off by one:** A. Item 0 doesn’t exist, so the first read is empty and the last item is missed — Scratch lists start at 1. Starting at 0 is an off-by-one error.
4. **Plan your code:** Goals: Counts 4 hot days; Uses a condition in the loop; Reports it.
5. **Trace in Python:** A. 2

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.